A PROJECT REPORT

on

"MULTILINGUAL EMOTION-AWARE SENTIMENT INTELLIGENCE SYSTEM FOR TEMPORAL ANALYSIS OF PUBLIC DISCOURSE"

Submitted to
KIIT Deemed to be University

In Partial Fulfillment of the Requirement for the Award of

BACHELOR'S DEGREE IN
INFORMATION TECHNOLOGY

BY

RAJTANU DASGUPTA	2206200
RICHIK DEY	2206203
SOUMYA BHATTACHARJEE	2206219
SUBHADEEP MONDAL	2206221

UNDER THE GUIDANCE OF
DR. PARTHA SARATHI PAUL

SCHOOL OF COMPUTER ENGINEERING
KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY
BHUBANESWAR, ODISHA - 751024

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KIIT Deemed to be University
School of Computer Engineering
Bhubaneswar, ODISHA - 751024

CERTIFICATE

This is to certify that the project entitled

"MULTILINGUAL EMOTION-AWARE SENTIMENT INTELLIGENCE SYSTEM FOR TEMPORAL ANALYSIS OF PUBLIC DISCOURSE"

submitted by

RAJTANU DASGUPTA	2206200
RICHIK DEY	2206203
SOUMYA BHATTACHARJEE	2206219
SUBHADEEP MONDAL	2206221

is a record of bonafide work carried out by them, in partial fulfillment of the requirement for the award of the Degree of Bachelor of Engineering (Information Technology) at KIIT Deemed to be University, Bhubaneswar. This work was done during the years 2025-2026, under my guidance.

Date: 09 / 04 / 2026

Dr. Partha Sarathi Paul
Project Guide

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Signature of Rajtanu Dasgupta………………..

Signature of Richik Dey…………………

Signature of Soumya Bhattacharjee………………..

Signature of Subhadeep Mondal………………..

ABSTRACT

The Russia-Ukraine conflict has emerged as one of the most significant geopolitical events of recent times, generating extensive discourse on social media platforms, particularly Twitter. The rapid growth of social media platforms has transformed public discourse into a rich and dynamic source of real-time information, especially during global conflicts. Understanding public sentiment and emotional patterns during such events is crucial for policymakers, researchers, and analysts. This project presents a comprehensive framework for multilingual emotion-aware sentiment analysis to study temporal changes in public discourse across different phases of a war.

The proposed system analyzes large-scale multilingual textual data collected from social media and categorizes it into three temporal phases: Early War, Mid War, and Later War. Advanced Natural Language Processing (NLP) techniques are applied to perform 3 sentiment classification (positive, negative, neutral) along with 6 fine-grained emotion detection (anger, fear, joy, sadness, love, and surprise). The study also incorporates engagement metrics such as retweets and likes to understand how different sentiments and emotions influence public interaction over time.

Experimental results reveal that negative sentiment consistently dominates across all phases, with a noticeable increase in intensity during later stages of the conflict. Emotions such as anger and sadness are found to be most prevalent, while engagement analysis shows that emotionally charged content tends to receive higher user interaction. Statistical tests further validate the significance of variations in sentiment and emotion across different phases.

This work highlights the importance of combining sentiment analysis with emotional intelligence and temporal context to gain deeper insights into public opinion. The proposed framework can be extended to other domains such as political analysis, crisis management, and social behavior monitoring, making it a valuable tool for large-scale data-driven decision-making.

Keywords: Sentiment Analysis, Emotion Detection, Multilingual NLP, Temporal Analysis, Social Media Analytics, War Discourse

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Chapter 1

Indroduction

In recent years, social media platforms have evolved into significant channels for expressing public opinion, particularly during global conflicts. During war situations, massive volumes of multilingual user-generated content are produced in real time, capturing diverse emotional reactions such as anger, fear, sadness, and hope. Analyzing these sentiments and emotions is crucial for understanding public perception, information flow, and the psychological impact of conflict across different stages.

However, existing sentiment analysis approaches are often limited to single-language datasets and focus primarily on basic polarity classification (positive, negative, neutral), overlooking deeper emotional insights. Furthermore, most traditional methods treat data as static, failing to capture the dynamic nature of public discourse over time. This creates a significant gap in understanding how sentiments and emotions evolve across different phases of prolonged events such as wars.

To address these challenges, this project proposes a **Multilingual Emotion-Aware Sentiment Intelligence System** for temporal analysis of public discourse across Early, Mid, and Later war phases. The system integrates sentiment classification, fine-grained emotion detection, and engagement metrics such as likes and retweets to provide a comprehensive and dynamic understanding of public reactions.

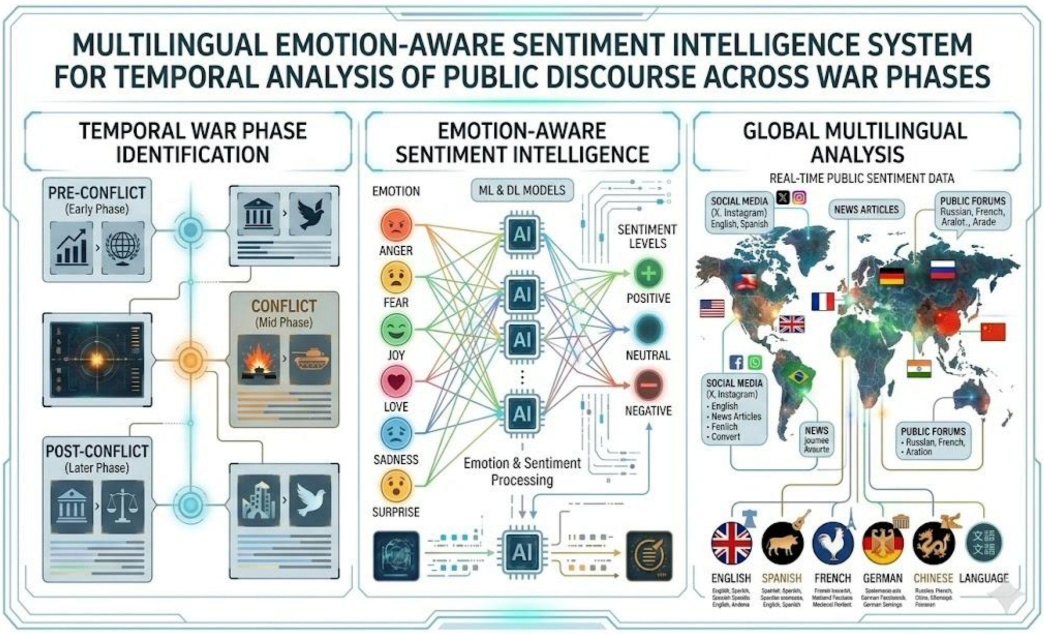


Figure 1.1: PROJECT ARCHITECTURAL OVERVIEW

1.1 Project Objectives

The primary objective of this project is to develop an intelligent and scalable system for analyzing multilingual public discourse by integrating sentiment analysis, emotion detection, and temporal dynamics across different phases of a war. The specific objectives of the project are as follows:

- **To perform multilingual sentiment analysis** on large-scale social media data and classify text into positive, negative, and neutral categories.
- **To detect fine-grained emotions** such as anger, fear, joy, sadness, love, and surprise, enabling a deeper understanding of public reactions beyond basic sentiment polarity.
- **To conduct temporal analysis** by dividing the dataset into Early War, Mid War, and Later War phases, and examining how sentiment and emotional patterns evolve over time.
- **To analyze user engagement patterns** (likes and retweets) and study their relationship with different sentiments and emotions.
- **To design a unified analytical framework** capable of handling multilingual data efficiently and consistently.
- **To apply statistical methods** such as Chi-Square tests and Cramér's V to validate the significance of observed patterns and relationships.
- **To visualize insights effectively** using graphs, charts, and trends for better interpretation and decision-making.
- **To provide meaningful interpretations** of results that can support real-world applications such as policy analysis, crisis monitoring, and social behavior understanding.

Overall, the project aims to bridge the gap between traditional sentiment analysis and real-world complex scenarios by incorporating emotion intelligence and temporal context into a single comprehensive system.

1.2 Report Structure

The report is organized as follows: Chapter 2 discusses the fundamental concepts and literature review, Chapter 3 presents the problem statement and system requirements, Chapter 4 describes the implementation and result analysis, Chapter 5 outlines the standards adopted, and Chapter 6 concludes the study with key insights and findings.

Chapter 2

Basic Concepts / Literature Review

In recent years, sentiment analysis and emotion detection have gained significant attention due to the rapid growth of social media platforms. Researchers have explored various techniques to understand public opinion by analyzing textual data generated during real-world events. This chapter presents the fundamental concepts and a review of existing approaches relevant to this project.

2.1 Sentiment Analysis

Sentiment analysis refers to the process of identifying the polarity of a given text, typically classified as positive, negative, or neutral. Traditional approaches relied on lexicon-based methods, where predefined dictionaries were used to assign sentiment scores to words. While simple and interpretable, these methods often fail to capture context and sarcasm.

With advancements in machine learning, supervised models such as Naïve Bayes, Support Vector Machines, and Logistic Regression have been widely used for sentiment classification. More recently, deep learning models like Recurrent Neural Networks (RNNs) and Transformer-based architectures have significantly improved accuracy by understanding contextual relationships in text.

2.2 Emotion Detection

Unlike sentiment analysis, which focuses on polarity, emotion detection aims to identify specific emotional states expressed in text, such as anger, fear, joy, sadness, love, and surprise. Emotion-aware analysis provides deeper insights into human reactions, especially during critical events like wars or disasters.

Existing research shows that combining sentiment and emotion analysis leads to more meaningful interpretations. However, emotion detection remains challenging due to language ambiguity and cultural differences in expression.

2.3 Multilingual Text Analysis

Most early studies in sentiment analysis focused on English datasets. However, social media data is inherently multilingual, especially in global events. Multilingual analysis involves processing text from different languages while maintaining consistency in classification.

Recent approaches use translation techniques or multilingual language models to address this issue. Despite progress, challenges such as language diversity, slang, and incomplete translations still affect performance.

2.4 Temporal Analysis of Social Media Data

Temporal analysis studies how sentiment and emotions change over time. In dynamic events like wars, public opinion is not static but evolves based on new developments.

Previous research has shown that analyzing data across different time intervals helps in identifying trends, sudden shifts, and patterns in public discourse. However, many existing systems do not integrate temporal segmentation effectively, limiting their ability to capture phase-wise variations.

2.5 Engagement Metrics Analysis

User engagement metrics such as likes and retweets play an important role in understanding the impact and reach of content. Highly engaging posts often reflect strong emotions or controversial opinions.

Studies indicate that negative or emotionally intense content tends to receive higher engagement. However, limited research has explored the relationship between sentiment, emotion, and engagement in a combined framework.

2.6 Research Gap

From the above discussion, the following gaps are identified:

- Most systems focus only on sentiment polarity and ignore detailed emotion analysis.
- Limited work has been done on multilingual datasets in a unified framework.
- Existing approaches often treat data as static, without considering temporal evolution.
- The relationship between sentiment, emotion, and user engagement is not deeply explored.

To overcome these limitations, this project proposes an integrated approach that combines multilingual sentiment analysis, emotion detection, and temporal analysis across different war phases. By incorporating engagement metrics and statistical validation, the system aims to provide a more comprehensive understanding of public discourse.

Chapter 3

Problem Statement / Requirement Specifications

In the modern digital era, social media platforms generate massive volumes of real-time data during global events such as wars. This data reflects public opinion, emotional reactions, and information spread across diverse populations. However, analyzing this data effectively presents several challenges.

Existing sentiment analysis systems primarily focus on single-language datasets and basic polarity classification (positive, negative, neutral). Such approaches fail to capture the complexity of human emotions expressed in text, especially during critical situations like war. Additionally, most systems treat data as static and do not analyze **temporal dynamics** - failing to capture how **public sentiment and emotional expressions** evolve over time.

Another major limitation is the lack of integration between sentiment, emotion, and user engagement metrics such as likes and retweets. Without combining these aspects, it becomes difficult to understand the true impact and influence of public discourse.

Furthermore, multilingual data introduces additional complexity due to language diversity, informal expressions, and translation inconsistencies. Current solutions do not adequately address these challenges in a unified framework.

Therefore, there is a need for a comprehensive system that can perform multilingual sentiment analysis, detect emotions, and analyze temporal variations across different phases of a war while incorporating engagement metrics for deeper insights.

3.1 Project Planning

The project is planned and executed in the following structured steps:

- **Data Collection:** Selection of datasets representing early, mid, and later war phases
- **Data Preprocessing:** Cleaning text, removing noise, handling missing values
- **Language Handling:** Processing multilingual data using unified models
- **Sentiment Analysis:** Classification into positive, negative, and neutral categories
- **Emotion Detection:** Identification of emotions such as anger, fear, sadness, and joy
- **Temporal Segmentation:** Dividing data into different war phases
- **Engagement Analysis:** Evaluating likes and retweets for impact measurement
- **Visualization:** Generating graphs and plots for better interpretation

- **Statistical Testing:** Applying Chi-square tests and Cramér's V to validate differences across phases

3.2 Project Analysis (SRS)

3.2.1 Functional Requirements

The system must:

- Accept multilingual text data as input
- Perform preprocessing and normalization of text
- Classify sentiment into predefined categories
- Detect emotions from textual content
- Segment data based on temporal phases
- Analyze engagement metrics such as likes and retweets
- Generate visual outputs such as graphs and charts
- Provide statistical analysis results
- Store and retrieve processed data efficiently

3.2.2 Non-Functional Requirements

The system should:

- Ensure **accuracy** in sentiment and emotion classification
- Maintain **scalability** for handling large datasets
- Provide **efficient** processing time
- Ensure **reliability** of outputs
- Maintain **readability and interpretability** of results
- Be **user-friendly** for analysis and visualization

3.3 System Design

3.3.1 Design Constraints

The project is developed under the following constraints:

- **Hardware Requirements:**

Components	Specifications
Processor	Intel Core i5/i7 or AMD Ryzen 5/7
RAM	Minimum 8GB (16GB Recommended)
Storage	At least 256GB SSD (512GB Recommended)
GPU	NVIDIA GTX 1660/RTX 2060 for deep learning acceleration
Operating System	Windows 10/11, Ubuntu 20.04+
Internet	Stable Connection for data collecton and model training

- **Software Requirements:**

Components	Technologies Used
Programming Language	Python (Latest Version)
Operating Systems	Windows, Ubuntu, macOS
Frameworks & Libraries	Pandas, Numpy, Matplotlib, Seaborn, Scikit-learn
Development Tools	Jupyter Notebook, Google Colab, VS Code

3.3.2 System Architecture / Block Diagram

The system follows a pipeline-based architecture consisting of the following stages:

- **Input Layer:** Raw multilingual social media data
- **Preprocessing Layer:** Cleaning, tokenization, normalization
- **Processing Layer:** Hugging Face Transformers classification
- **Temporal Analysis Layer:** Division into early, mid, and later phases
- **Engagement Analysis Layer:** Likes and retweets evaluation per sentiment/emotion
- **Visualization Layer:** Graphs and plots generation
- **Output Layer:** Final insights and statistical results

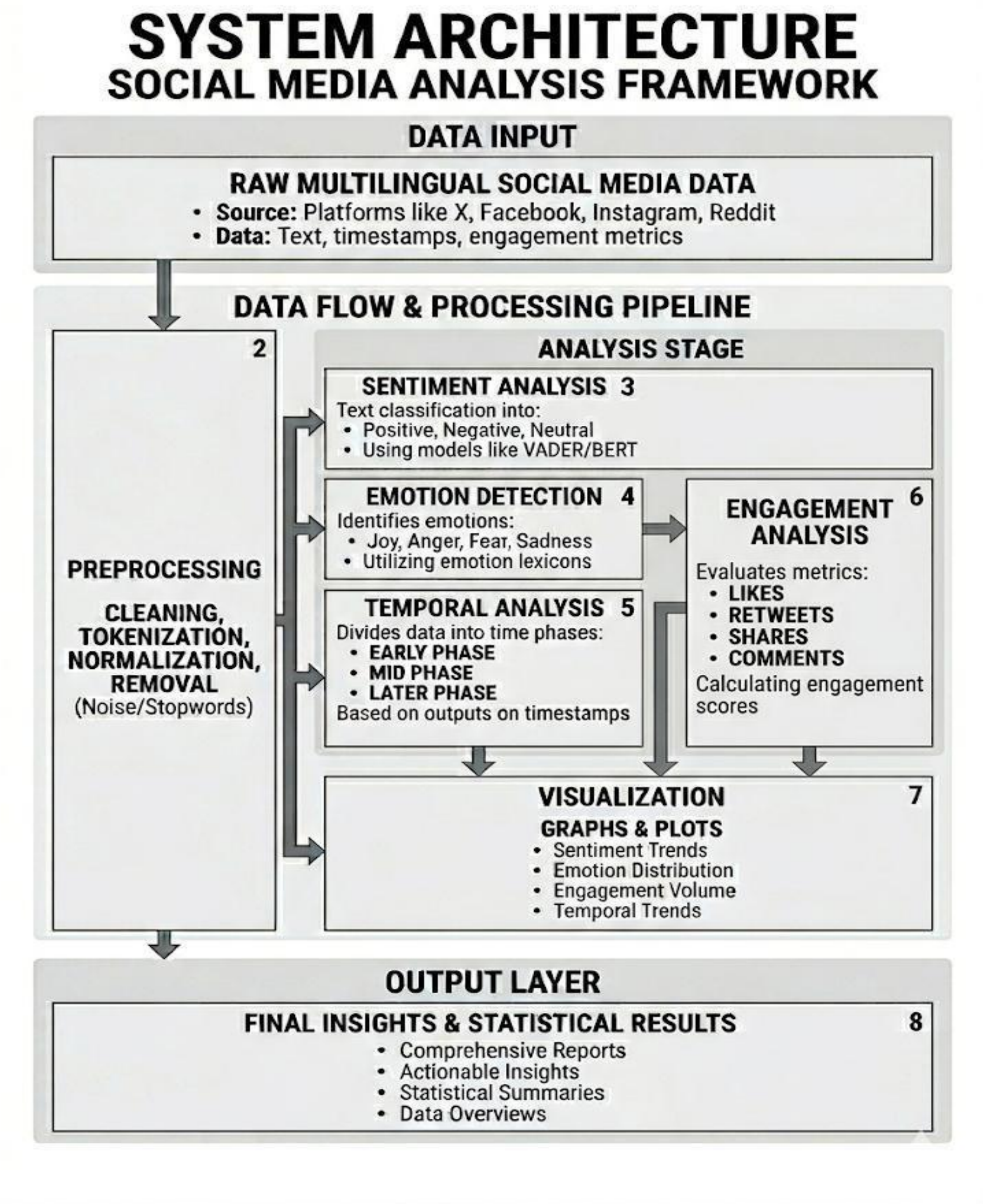


Figure 3.1: SYSTEM BLOCK DIAGRAM

Features of Deployed Model:

✔ Multilingual Language Detection

✔ Automatic Translation

✔ VADER Sentiment Analysis

✔ BERT Emotion Detection

✔ Professional UI for Research Demo

✔ Real-time Analysis

Chapter 4

Implementation

4.1 Methodology / Proposal

The proposed system follows a structured pipeline to perform multilingual emotion-aware sentiment analysis across different war phases. The methodology is designed to ensure accurate data processing, meaningful feature extraction, and reliable analysis of temporal trends.

4.1.1 Data Collection

The implementation begins with data collection, where tweets corresponding to early (47994 raw tweets), mid (32046 raw tweets), and later (34970 raw tweets) war phases are gathered. The dataset is carefully divided into three balanced segments to avoid bias and ensure fair comparison across phases.

Resource (Kaggle): "*https://www.kaggle.com/datasets/bwandowando/ukraine-russian-crisis-twitter-dataset-1-2-m-rows*"

4.1.2 Data Preprocessing

In the data preprocessing stage, raw text data is cleaned by removing URLs, mentions, hashtags, special characters, and stop words. Text normalization techniques such as lowercasing and tokenization are applied. For multilingual handling, language normalization and encoding standardization are performed to maintain consistency across different languages.

4.1.3 Feature Extraction

Next, feature extraction is carried out using NLP techniques. Text data is transformed into numerical representations using methods such as TF-IDF and BoW vectorization. These features are used as inputs for sentiment and emotion classification models.

4.1.4 Sentiment and Emotion Analysis

The sentiment analysis module classifies tweets into positive, negative, and neutral categories. Alongside this, an emotion detection module identifies emotional states such as anger, fear, sadness, joy, and surprise from Hugging Face classification pipeline. This dual-layer analysis provides deeper insight compared to traditional sentiment-only approaches.

- **For English Tweets:**

Sentiment: VADER (lexicon-based, optimized for social media)

Emotion: *"bhadresh-savani/bert-base-uncased-emotion"* (BERT fine-tuned on 6 emotions)

- **For Multilingual Tweets:**

Sentiment: *"cardiffnlp/twitter-xlm-roberta-base-sentiment"* (multilingual XLM-RoBERTa)

Emotion: *"MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7"* (mDeBERTa multilingual)

4.1.5 Model Training and Evaluation:

(a) Machine Learning Models Used: Linear Regression, Logistic Regression, Naive Bayes (Multinomial), SVM (Linear Support Vector Machine), Decision Tree, Random Forest, KNN (K-Nearest Neighbor), XGBoost (Extreme Gradient Boosting).

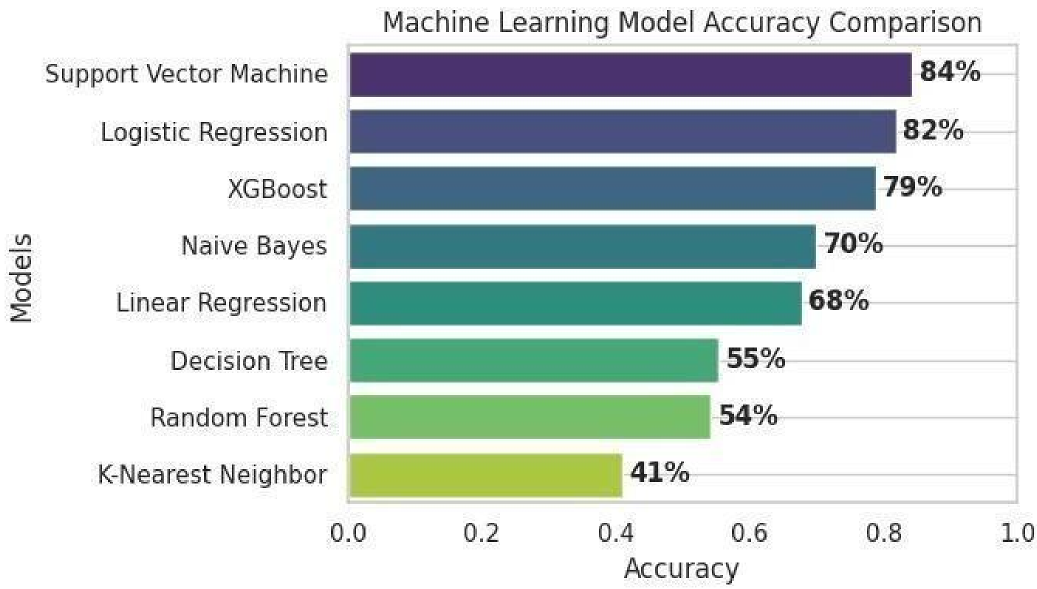


Figure 4.1: MACHINE LEARNING MODELACCURACYCOMPARISON

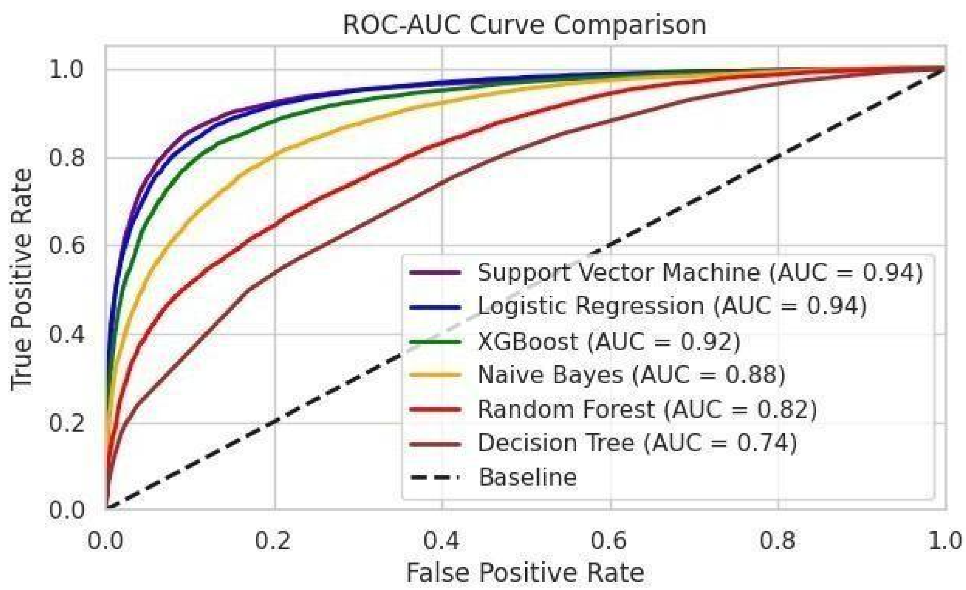


Figure 4.2: ROC-AUC CURVE COMPARISON FOR MLMODELS

(b) Deep Learning Models Used: MLP (Multi-Layer Perceptron), CNN (Convolutional Neural Network), RNN (Recurrent Neural Network), LSTM (Long Short-Term Memory), Transformer, Hybrid (CNN + LSTM), GAN (Generative Adversarial Network), BERT (Bidirectional Encoder Representations from Transformers).

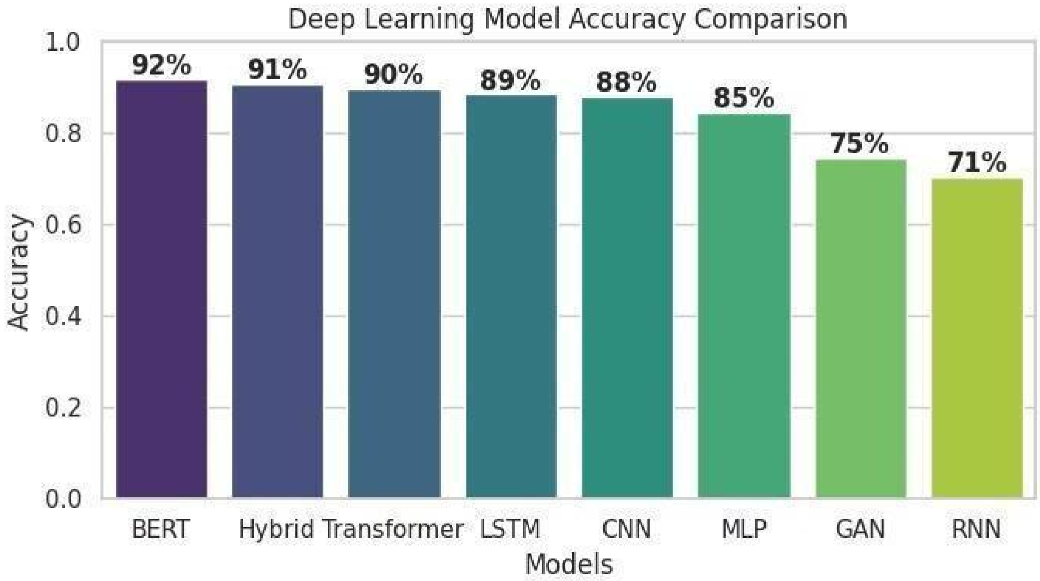
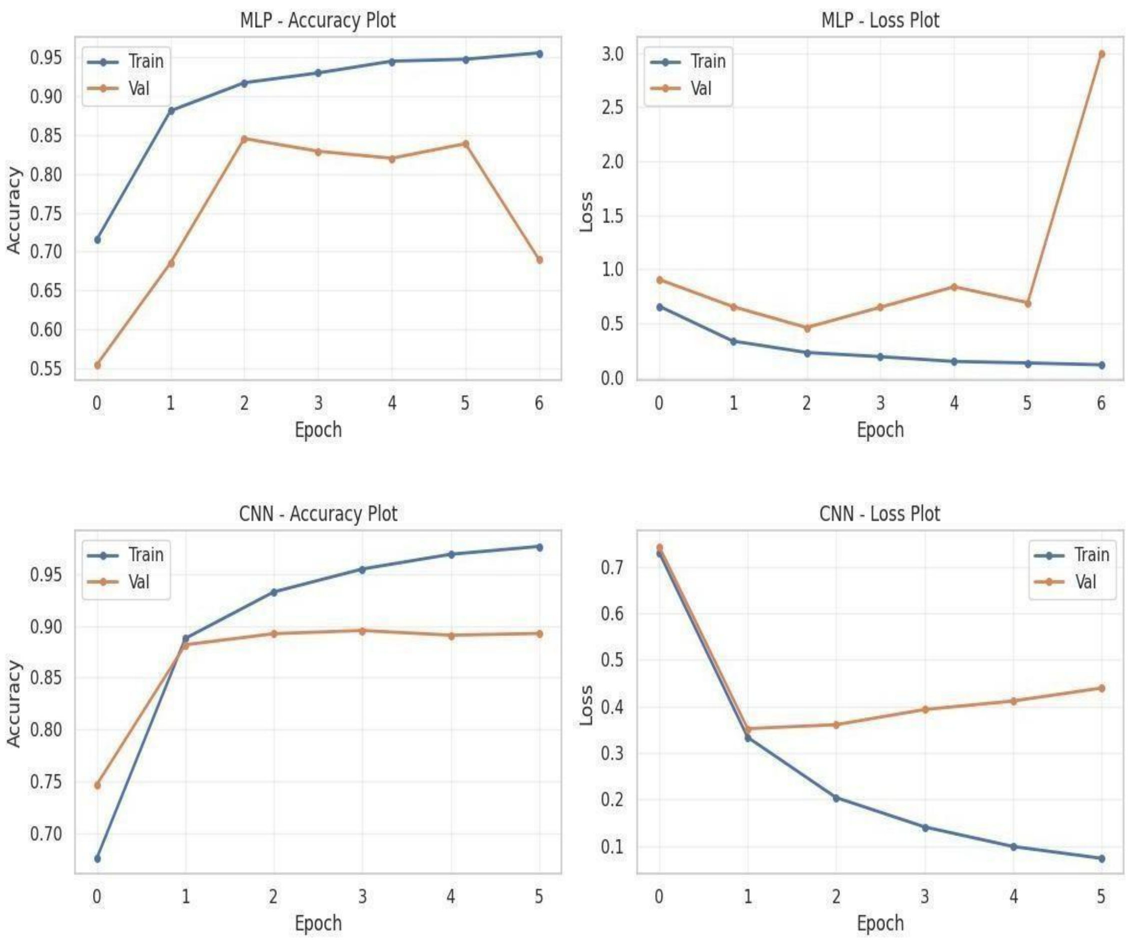


Figure 4.3: DEEP LEARNING MODEL ACCURACY COMPARISON



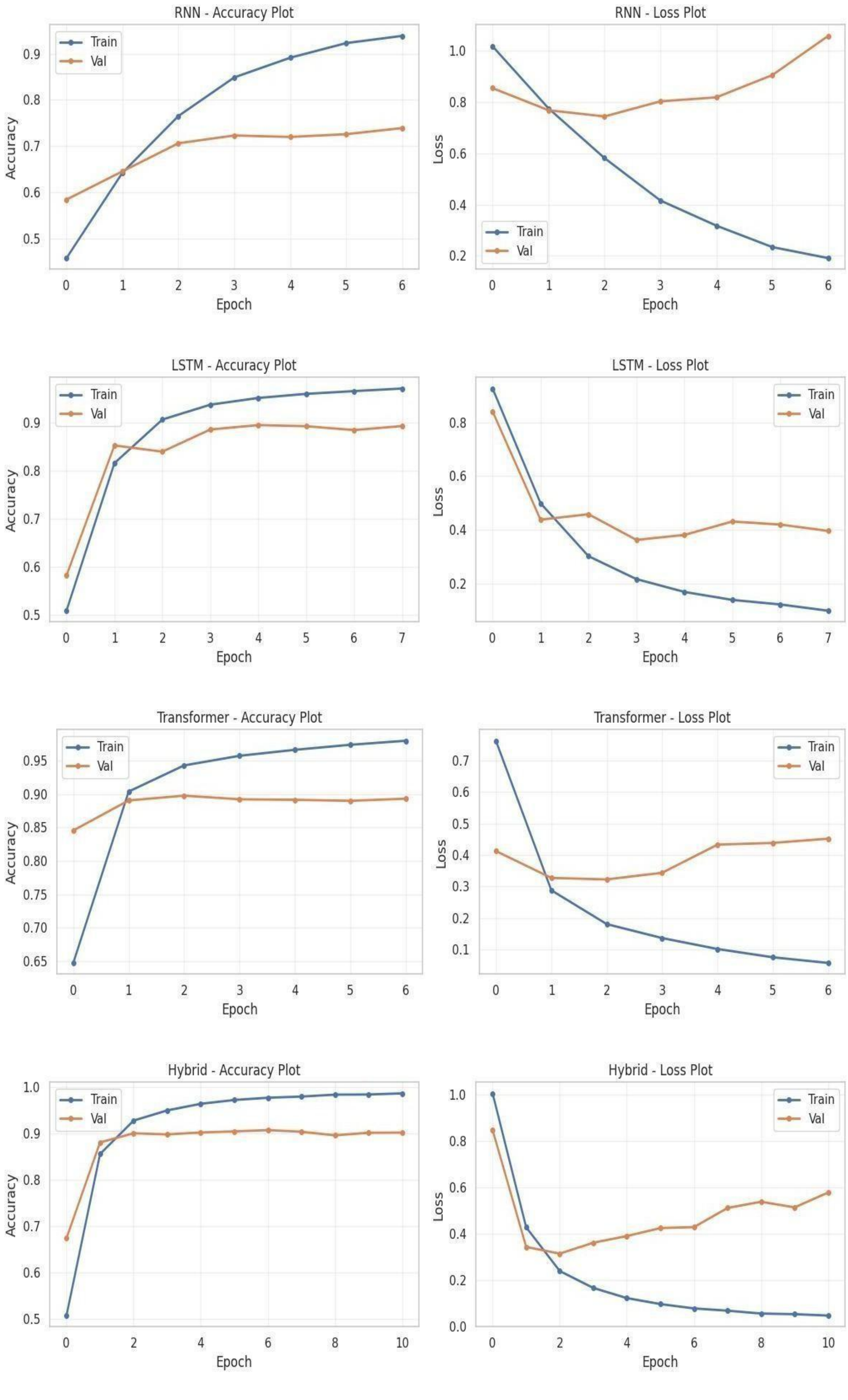


Figure 4.4: ACCURACY ANDLOSSPLOTFOR DL MODELS

4.1.6 Universal Multilingual Emotion and Sentiment Analyzer

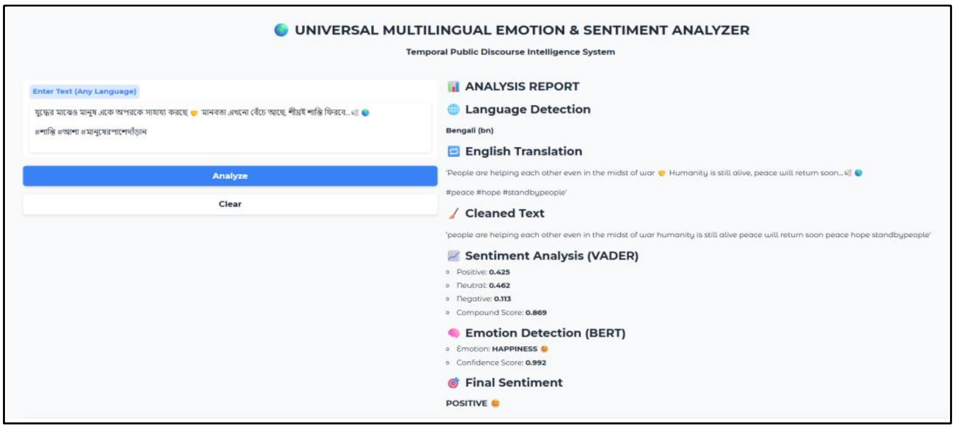


Figure 4.5: MULTILINGUAL EMOTION-SENTIMENT ANALYZER

4.1.7 Engagement Analysis

Further, engagement analysis is performed using metrics such as likes, retweets, and replies. These metrics help in understanding how different sentiments and emotions influence user interaction.

4.1.8 Temporal Analysis

Finally, temporal analysis is conducted by comparing outputs across early, mid, and later phases. Graphs and visualizations are generated to identify trends, variations, and patterns in public discourse over time.

4.2 Testing / Verification Plan

To validate the correctness and reliability of the system, multiple test cases were designed and executed. The system was tested for data processing accuracy, classification correctness, and output visualization:

Test ID	Test Case Title	Test Condition	System Behavior	Expected Result
T01	Dataset Load	Valid CSV file	Loads successfully	Data displayed
T02	Data Preprocess Check	Raw tweet input	Cleans and tokenizes text	Cleaned and structured text
T03	Sentiment Classification	Input text	Classified correctly	Valid sentiment
T04	Emotion Detection	Input text	Emotion assigned	Accurate emotion
T05	Engagement Analysis	Tweet metadata input	Calculates metrics	Correct likes/retweets
T06	Temporal Analysis	Phase-wise dataset	Generates comparative graphs	Clear trend viz. outputs
T07	Statistical Test	Valid input	Chi-square computed	Valid p-value

4.3 Results and Analysis

The results obtained from the implementation provide meaningful insights into public sentiment and emotional behavior across different war phases:

4.3.1 Sentiment Trends

- Negative sentiment dominates across all phases
- Increase from 62% (Early) → 68% (Later)
- Indicates rising tension over time

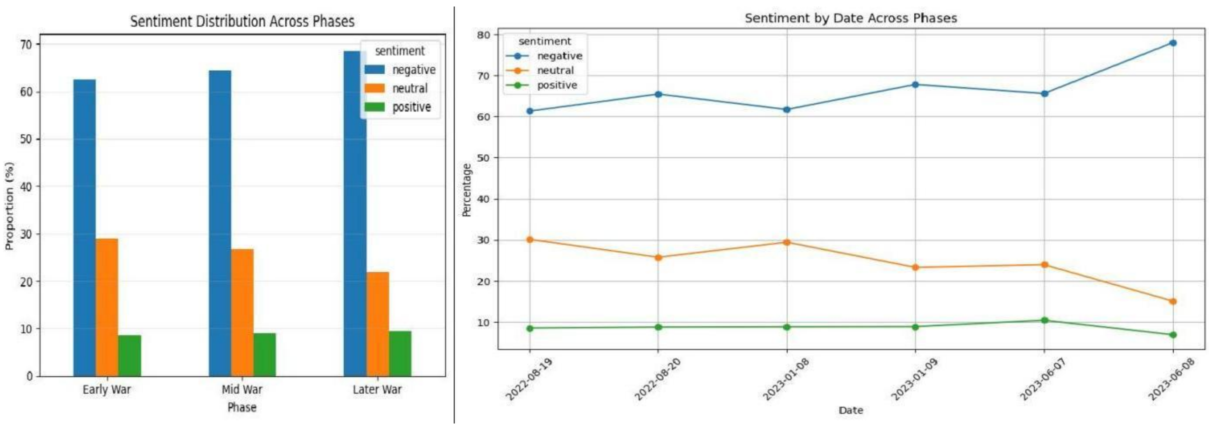


Figure 4.6: SENTIMENT DISTRIBUTION ACROSSPHASES

4.3.2 Emotion Trends

- Anger is the most dominant emotion (~50%)
- Fear increases in later stages
- Joy and love remain minimal

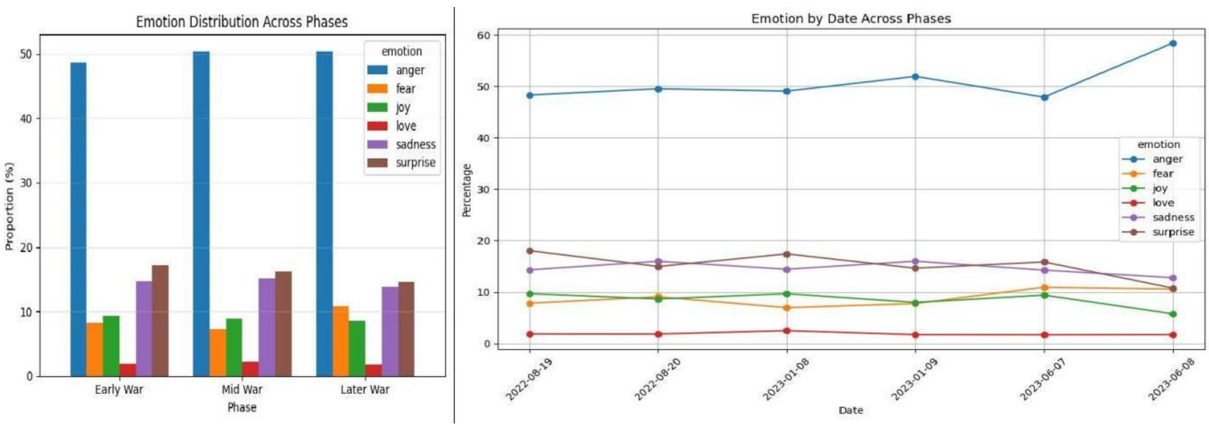


Figure 4.7: EMOTIONDISTRIBUTION ACROSS PHASES

4.3.3 Retweets and Sentiment Trends

- Negative sentiment receives the highest retweets in Early and Mid phases (~66%), indicating strong amplification of negative content
- Later phase shows a shift, with neutral sentiment gaining the largest share of retweets (~44%), surpassing negative sentiment
- Positive sentiment retweets increase gradually over time (11% → 12% → 18%), suggesting slightly improved engagement with positive content

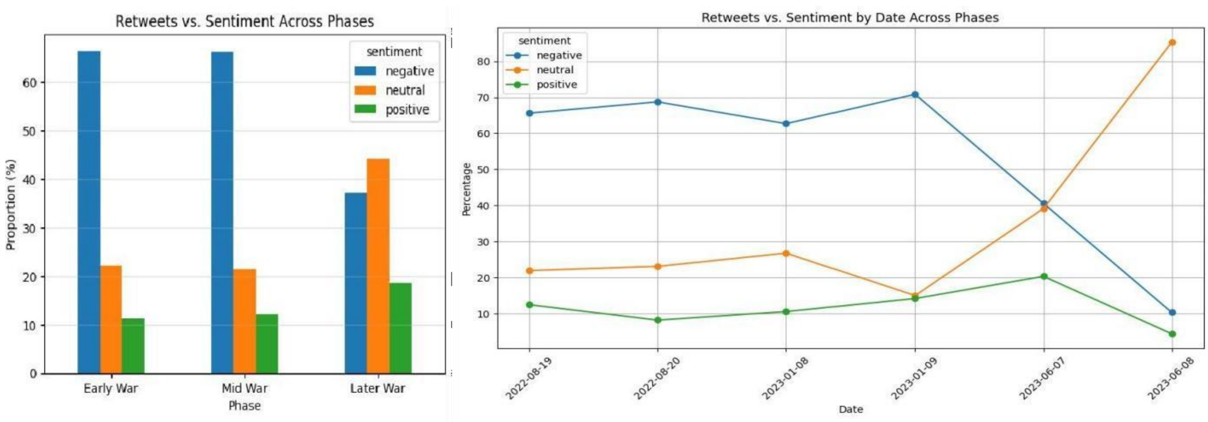


Figure 4.8: RETWEETS VS. SENTIMENT ACROSS PHASES

4.3.4 Likes and Sentiment Trends

- Negative sentiment receives the majority of likes in Early (60.55%) and Mid (63.71%) phases, showing stronger engagement with negative content initially
- Later phase shows a major shift, with neutral sentiment dominating likes (~68.64%), indicating reduced preference for negative posts
- Positive sentiment likes remain relatively stable in Early and Mid (~14–15%) but drop sharply in the Later phase (~4.78%), reflecting lower engagement with positive content

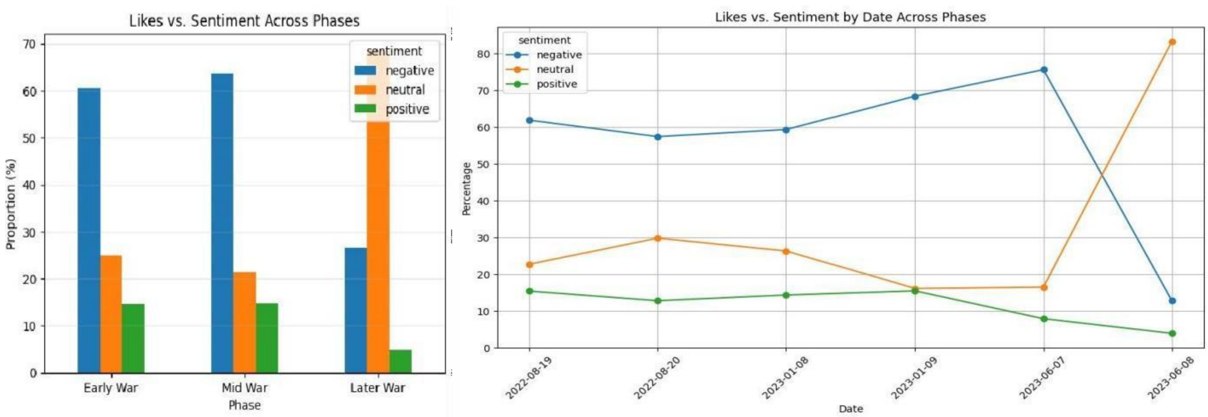


Figure 4.9: LIKESVS. SENTIMENTACROSSPHASES

4.3.5 Retweets and Emotion Trends

- Anger dominates retweets in Early (43.42%) and Mid (49.78%) phases, indicating strong amplification of anger-driven content

- In the Later phase, surprise becomes the most retweeted emotion (~33.53%), while anger decreases significantly (~25.66%)

- Joy and fear retweets increase in the Later phase, whereas love remains consistently minimal across all phases

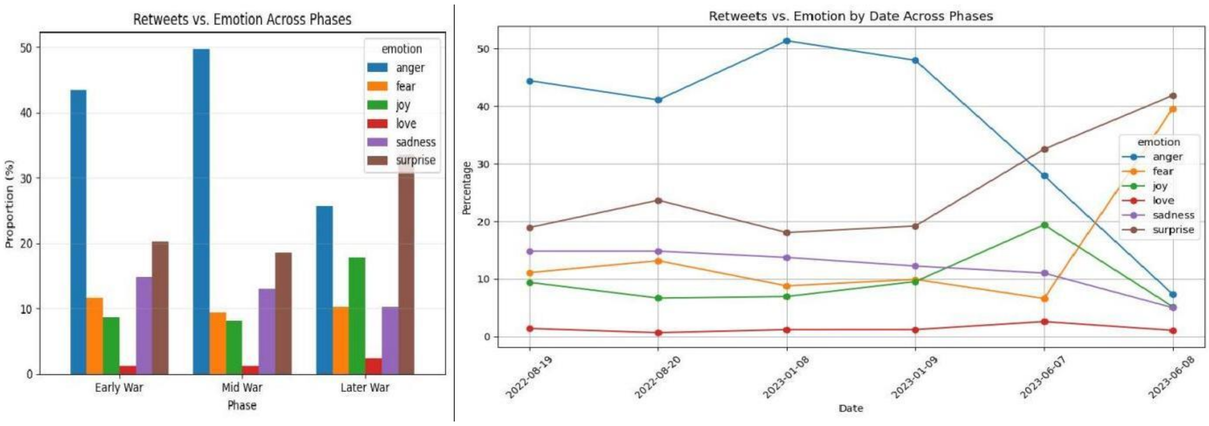


Figure 4.10: RETWEETS VS. EMOTION ACROSS PHASES

4.3.6 Likes and Emotion Trends

- Anger receives the highest likes in Early (43.90%) and Mid (48.97%) phases, showing strong engagement with anger-related content

- In the Later phase, surprise (35.55%) and fear (33.78%) dominate likes, while anger decreases significantly (~16.69%)

- Joy, sadness, and especially love remain comparatively low across all phases, indicating limited user preference for positive emotions

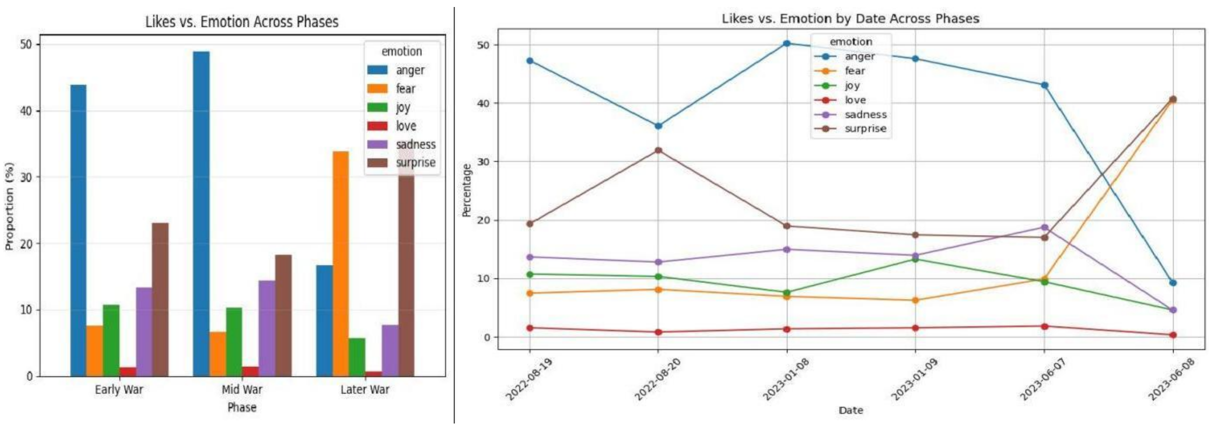


Figure 4.11: LIKESVS. EMOTIONACROSSPHASES

4.3.7 Engagement Analysis

(a) Sentiment and Emotion Relationship of Early Phase

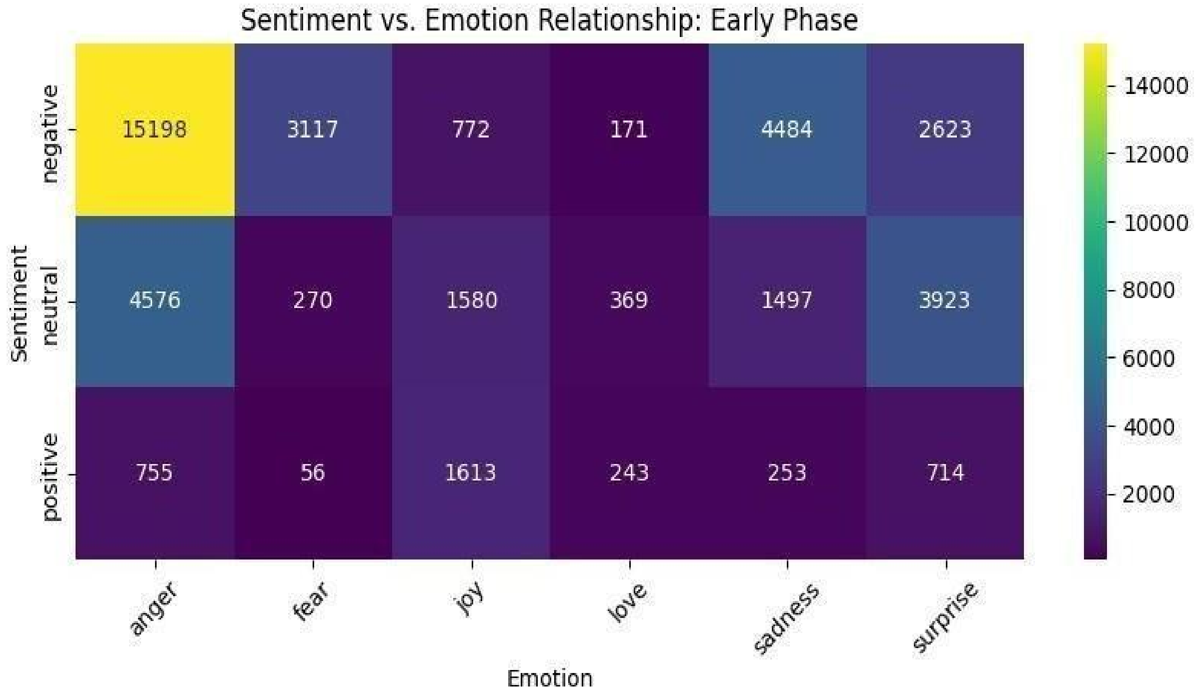


Figure 4.12: SENTIMENTVS. EMOTION RELATIONSHIP: EARLY PHASE

(b) Sentiment and Emotion Relationship of Mid Phase

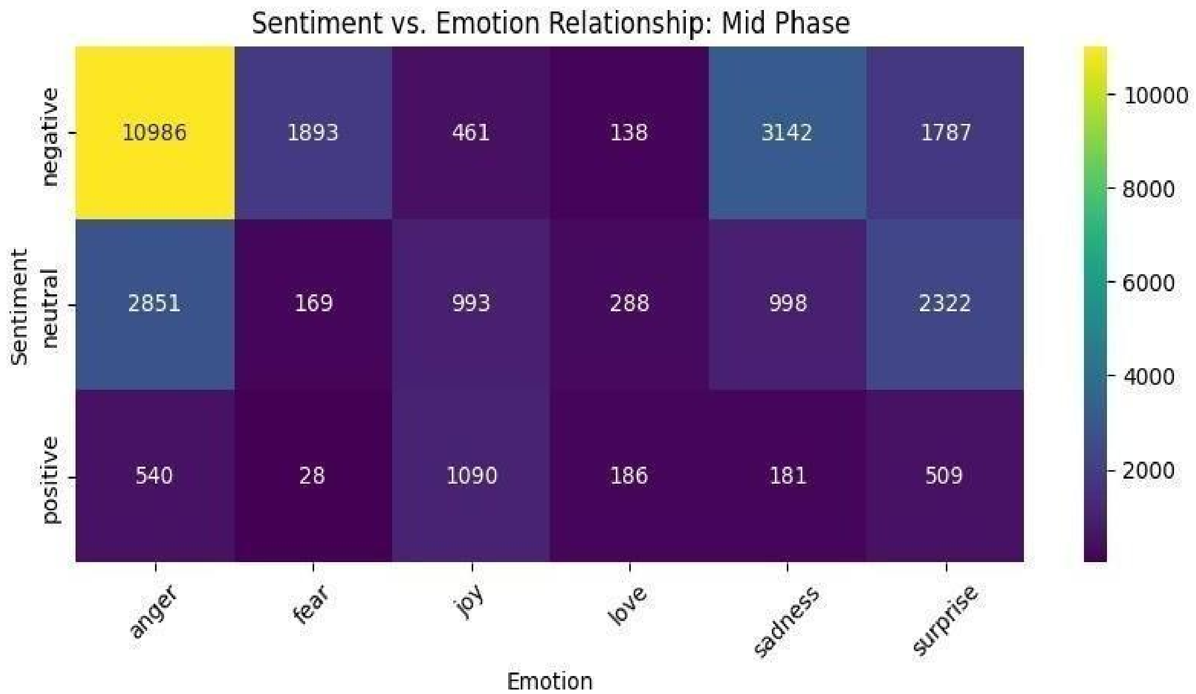


Figure 4.13: SENTIMENT VS. EMOTIONRELATIONSHIP: MID PHASE

(c) Sentiment and Emotion Relationship of Later Phase

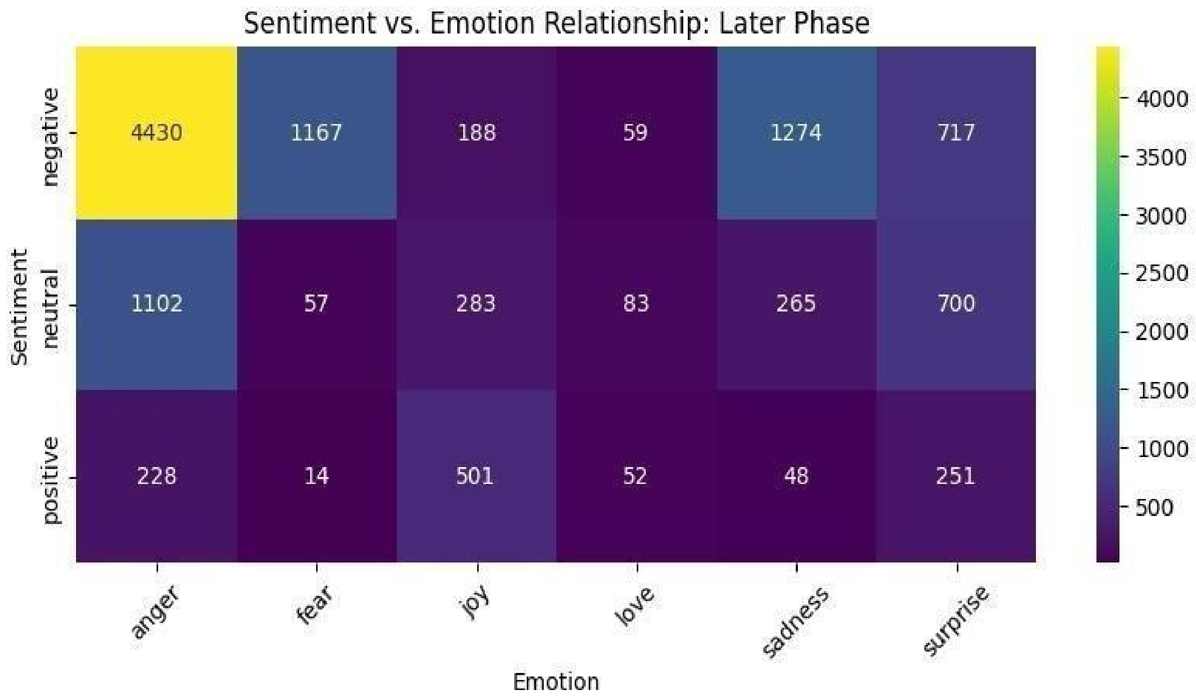


Figure 4.14: SENTIMENTVS. EMOTION RELATIONSHIP: LATER PHASE

(d) Sentiment and Emotion Relationship of All Phases

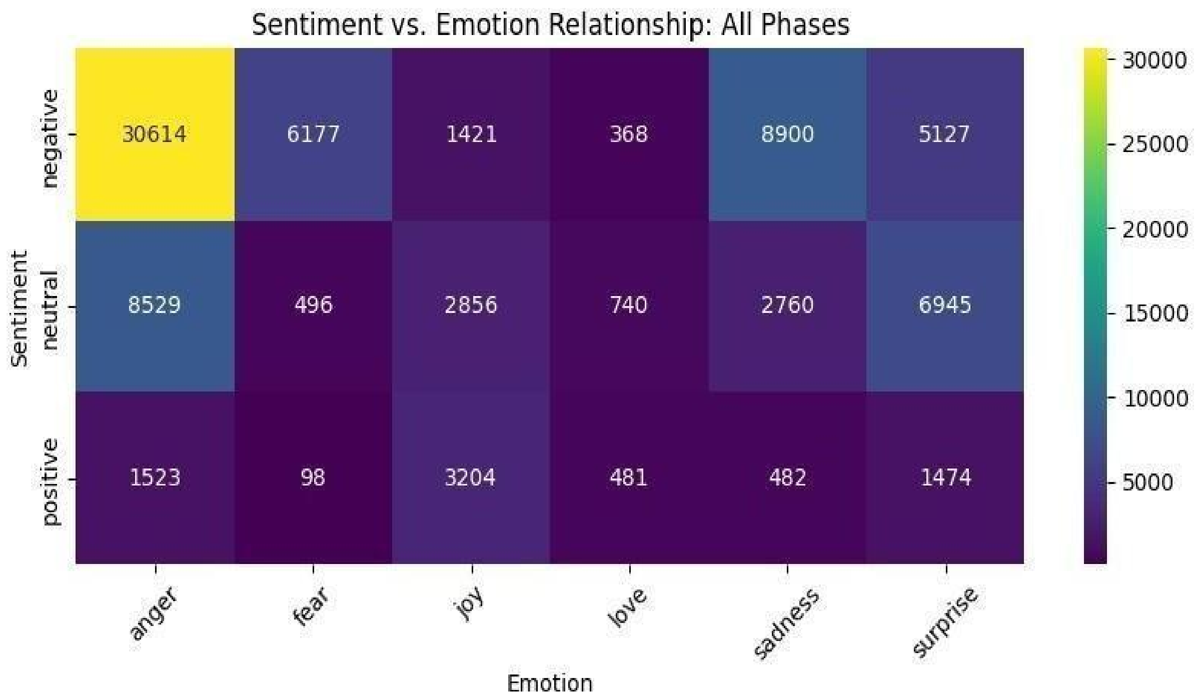


Figure 4.15: SENTIMENTVS. EMOTION RELATIONSHIP: ALL PHASES

(e) Average Retweets by Sentiment Over Time Across Phases

- Positive sentiment shows higher average retweets in Early (2.08) and Mid (2.32) phases, indicating stronger amplification compared to negative and neutral content

- Later phase exhibits a sharp increase in engagement, with neutral (116.35) and positive (111.20) sentiments receiving significantly higher average retweets

- Negative sentiment maintains relatively low average retweets across all phases, though it increases slightly in the Later phase (~31.13)

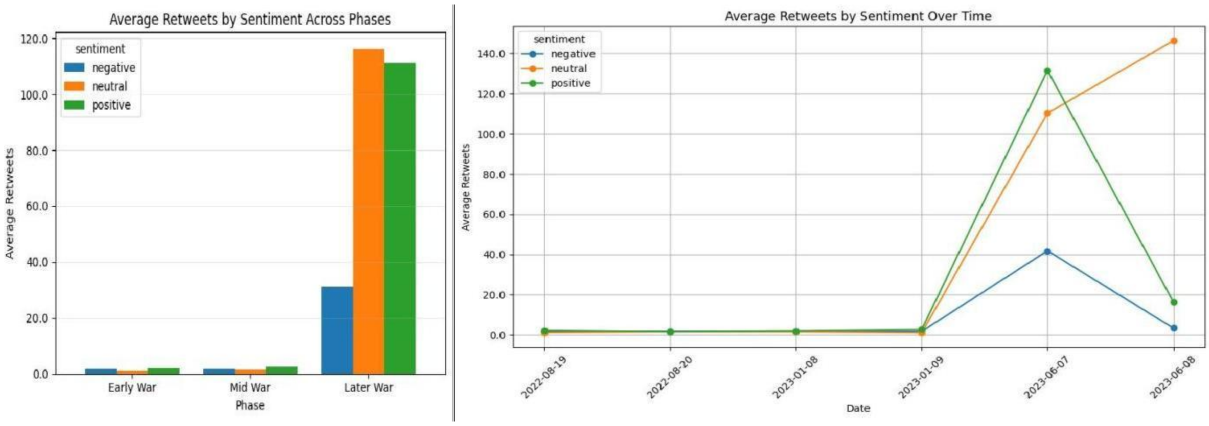


Figure 4.16: AVERAGE RETWEETS BY SENTIMENTACROSS PHASES

(f) Average Likes by Sentiment Over Time Across Phases

- Positive sentiment receives the highest average likes in Early (13.35) and Mid (14.96) phases, indicating stronger user appreciation for positive content initially

- Later phase shows a major shift, with neutral sentiment gaining the highest average likes (~25.05), surpassing both negative and positive sentiments

- Negative and positive sentiments experience a decline in average likes during the Later phase, suggesting reduced engagement compared to neutral content

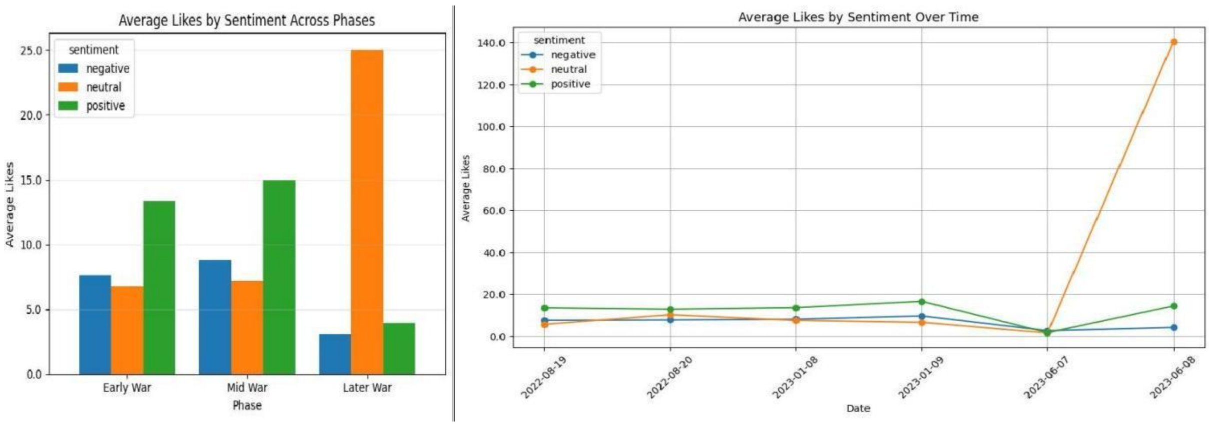


Figure 4.17: AVERAGE LIKESBYSENTIMENTACROSSPHASES

(g) Average Retweets by Emotion Over Time Across Phases

- Fear records the highest average retweets in Early (2.29) and Mid (2.14) phases, indicating stronger amplification of fear-related content initially

- Later phase shows a dramatic increase in engagement, with surprise (~131.70) and joy (~120.12) receiving the highest average retweets

- Anger and sadness maintain comparatively lower average retweets, while love remains the least engaged emotion in Early and Mid phases but rises in the Later phase

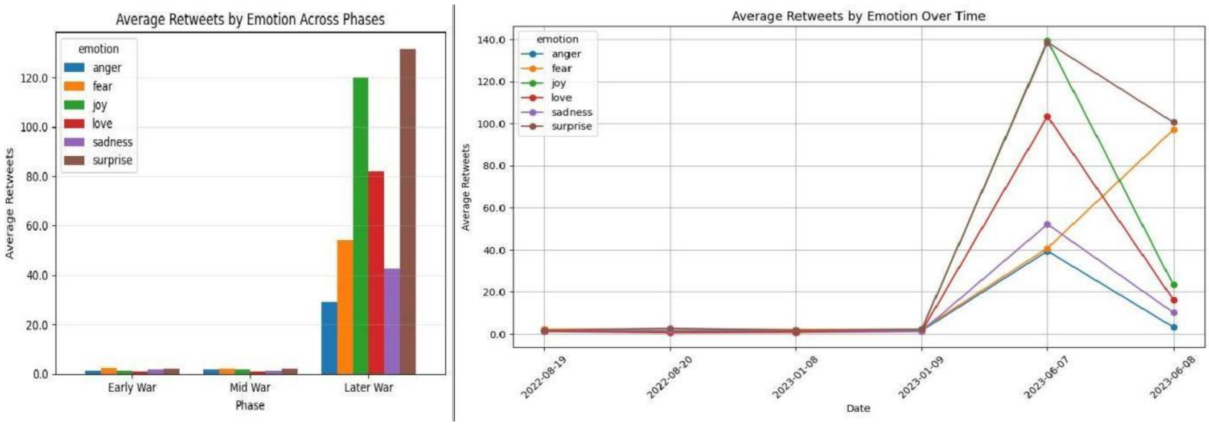


Figure 4.18: AVERAGE RETWEETS BY EMOTION ACROSS PHASES

(h) Average Likes by Emotion Over Time Across Phases

- Surprise receives the highest average likes in Early (10.59) and remains high in Mid phase (10.07), indicating strong user engagement with unexpected events

- Later phase shows a major shift, with fear (~24.80) and surprise (~19.37) dominating average likes, while other emotions decline

- Anger, sadness, and love maintain comparatively lower average likes across phases, with overall engagement dropping for most emotions in the Later phase

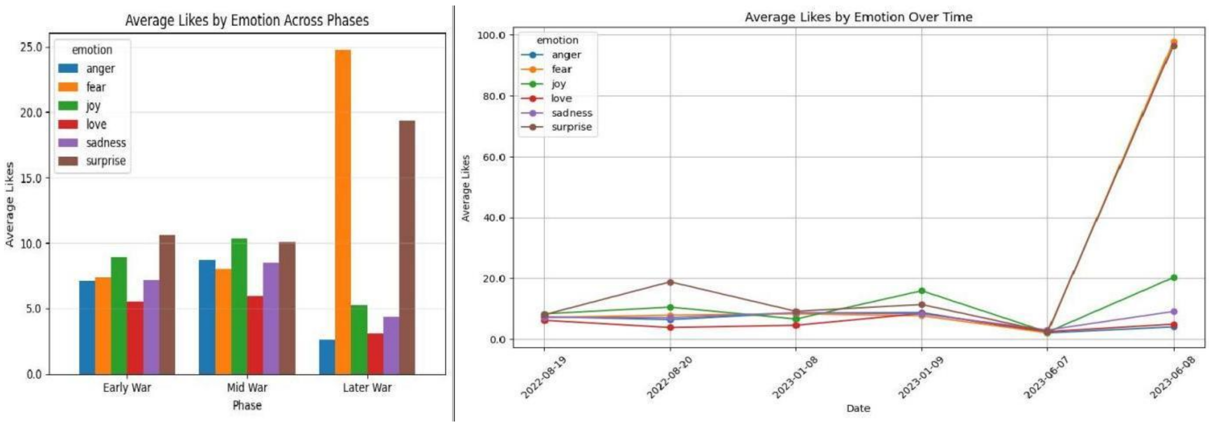


Figure 4.19: AVERAGE LIKES BY EMOTION ACROSS PHASES

4.3.8 Temporal Analysis

- During the **early phase**, the analysis shows a higher presence of fear and uncertainty, with a mix of neutral and negative sentiments. Engagement levels are moderate, indicating the initial reaction of users.

- In the **mid phase**, there is a noticeable increase in negative sentiment and emotions such as anger and frustration. This phase also shows higher engagement levels, suggesting active public participation and discussion.

- In the **later phase**, sentiment patterns stabilize, with a mix of neutral and slightly positive responses. Emotional intensity reduces compared to earlier phases, indicating adaptation or fatigue among users.

The combined temporal analysis (82195 clean tweets) highlights a clear evolution of public discourse. Graphical representations such as sentiment distribution charts, emotion frequency plots, and engagement trend graphs provide visual validation of these observations.

4.3.9 Statistical Validation

- Chi-square test shows significant relationship ($p < 0.05$)

- However, Cramer's V (~0.03) indicates weak effect size

- Differences exist, but changes are gradual rather than drastic

These results demonstrate the effectiveness of the proposed system in capturing both emotional depth and temporal dynamics.

4.3.10 Research Perspective

- Comparative temporal analysis across early, mid, and later phases shows limited variation in emotion distribution over time.

- Contrary to expected behavioral trends, dominant emotions such as anger did not significantly decrease across phases.

- Statistical observation indicates relative stability in emotional proportions, suggesting persistent public sentiment.

- Minimal fluctuation implies emotional consistency despite evolving contextual events.

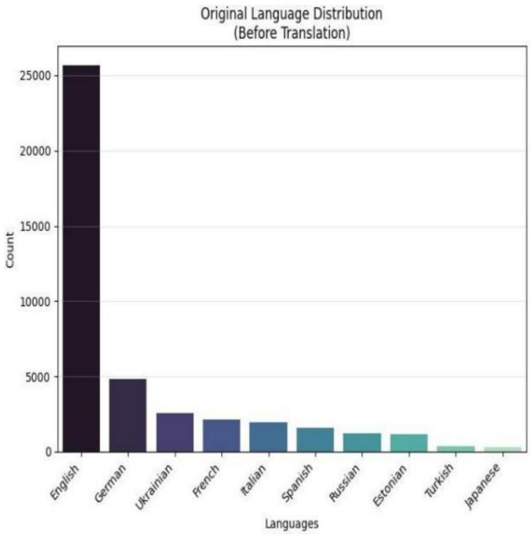
- Engagement metrics also support steady emotional intensity across temporal segments.

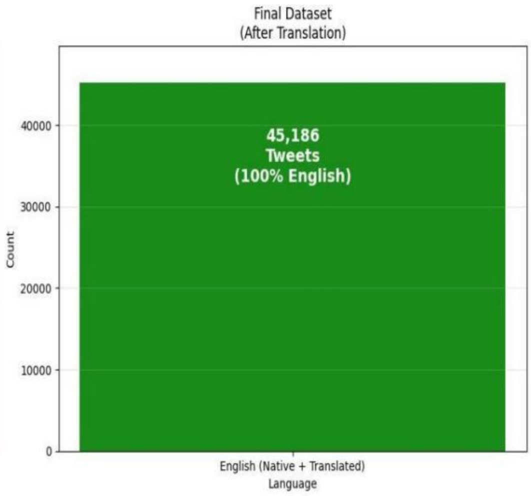
- This indicates that public discourse remains sentimentally polarized throughout the conflict timeline.

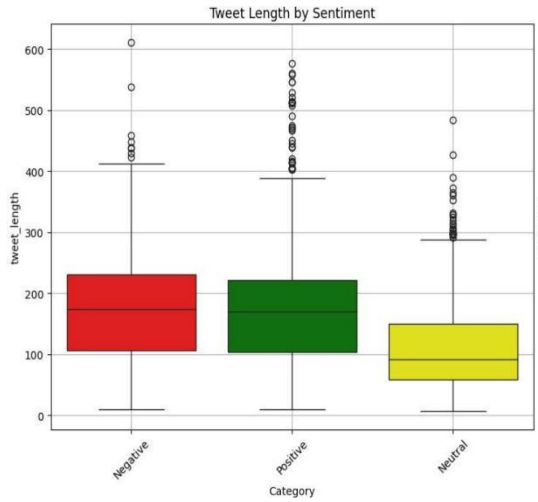
- The findings highlight weak temporal sensitivity in emotional transitions within the dataset.

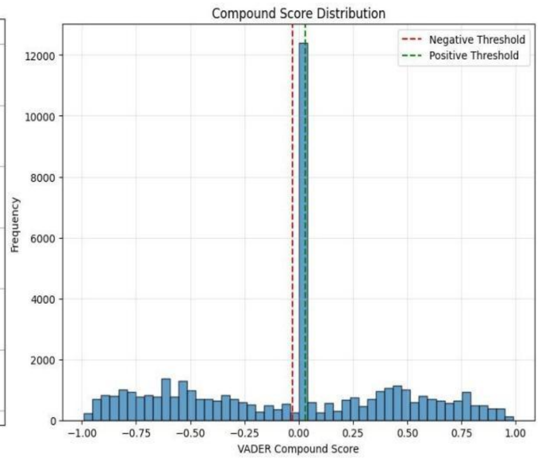
- Overall, results demonstrate low emotion fluctuation across phases, revealing sustained emotional response rather than gradual normalization.

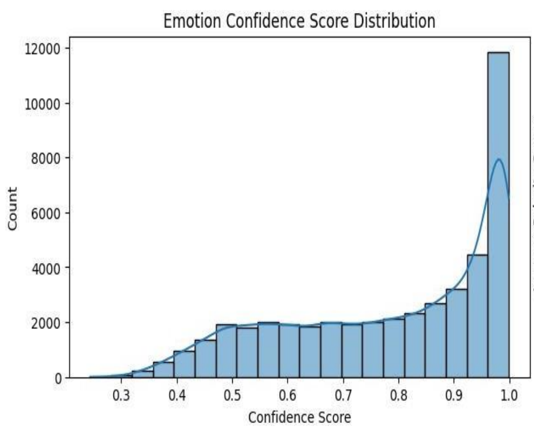
4.3.11 Some More Vizualisations

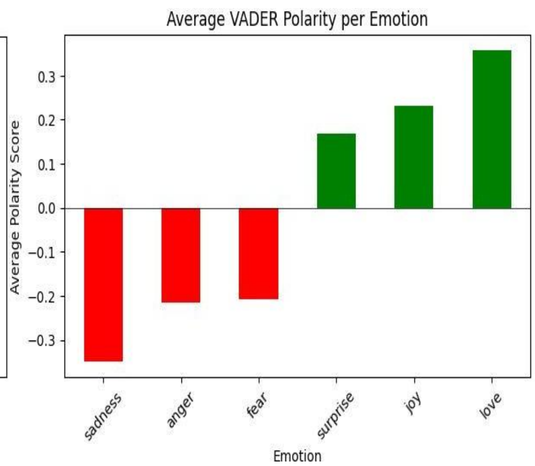


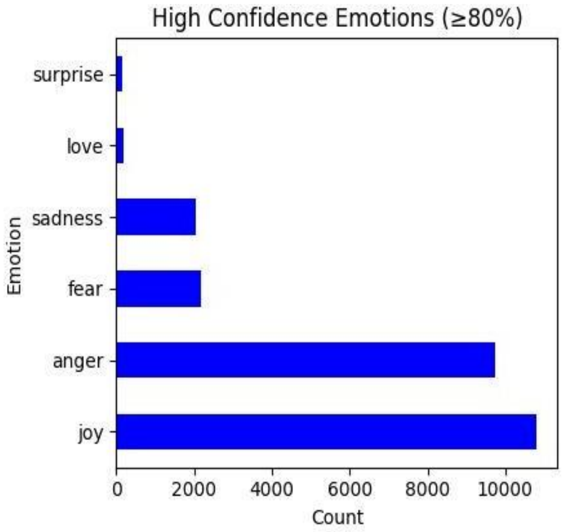


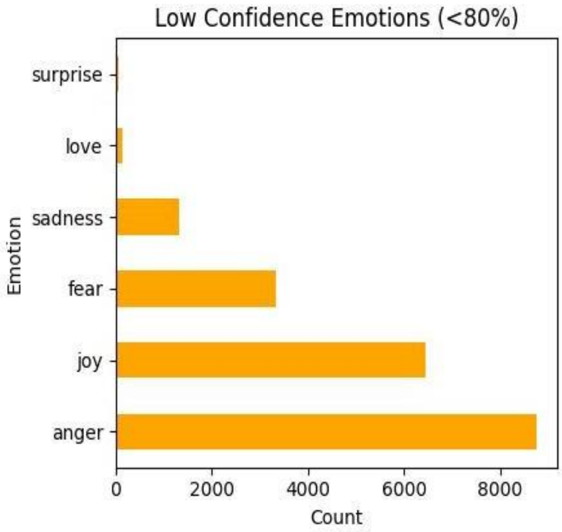


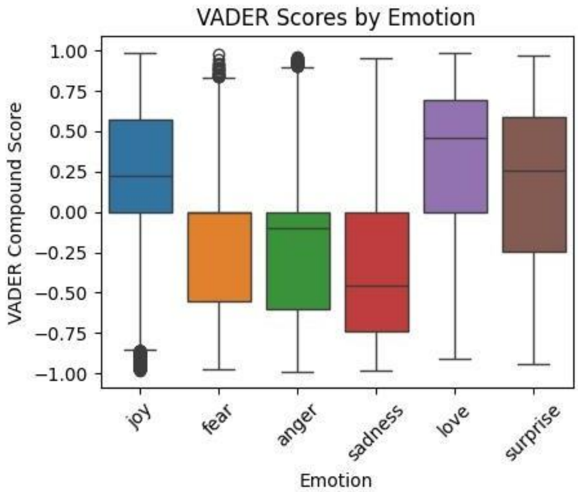


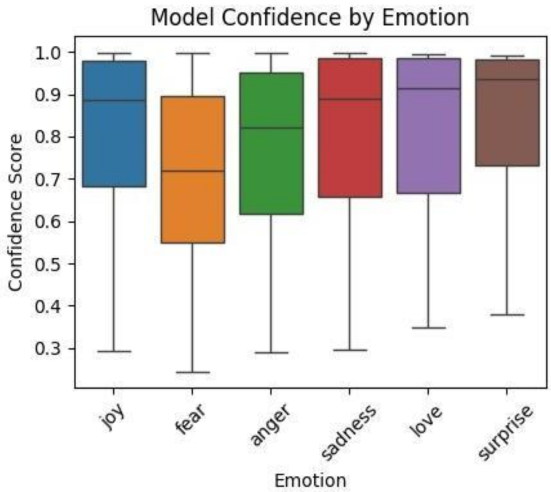


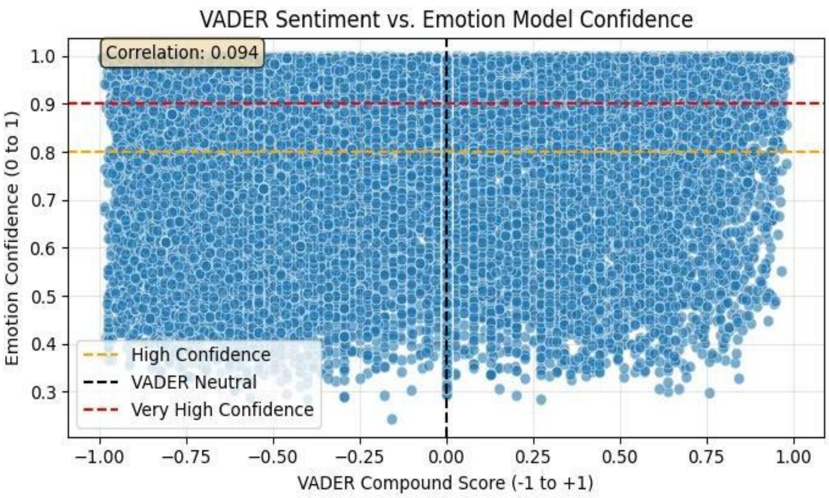


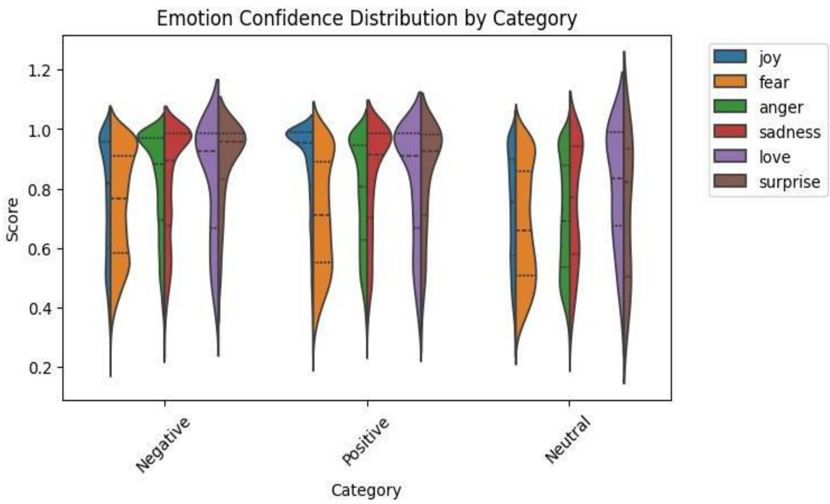

High Confidence Emotions (≥80%)
surprise
love
sadness
fear
anger
joy
Emotion
0
2000
4000
6000
8000
10000
Count


Low Confidence Emotions (<80%)
surprise
love
sadness
fear
joy
anger
Emotion
0
2000
4000
6000
8000
Count


VADER Scores by Emotion
VADER Compound Score
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
joy
fear
anger
sadness
love
surprise
Emotion


Model Confidence by Emotion
Confidence Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
joy
fear
anger
sadness
love
surprise
Emotion


VADER Sentiment vs. Emotion Model Confidence
Correlation: 0.094
Emotion Confidence (0 to 1)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
High Confidence
VADER Neutral
Very High Confidence
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
VADER Compound Score (-1 to +1)



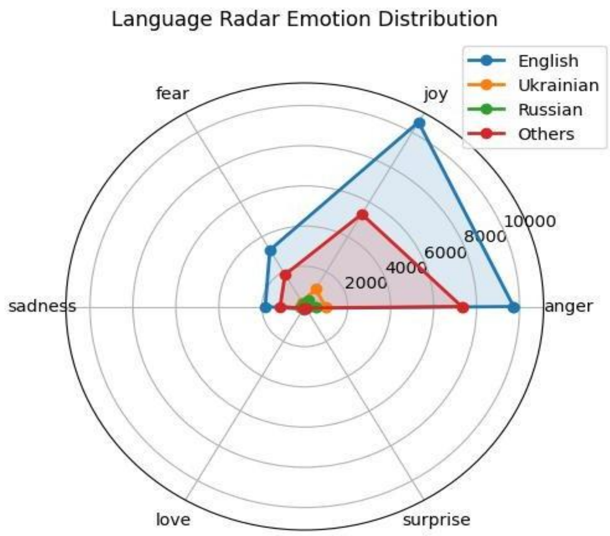


Figure 4.20: SOME MORE VIZUALISATIONS

	SentimentCount			Proportion(%)		
sentiment	negative	neutral	positive	negative	neutral	positive
phase						
Early War	26365	12215	3634	62.46	28.94	8.61
Mid War	18407	7621	2534	64.45	26.68	8.87
Later War	7835	2490	1094	68.61	21.81	9.58

	EmotionCount						Proportion(%)					
emotion	anger	fear	joy	love	sadness	surprise	anger	fear	joy	love	sadness	surprise
phase												
Early War	20529	3443	3965	783	6234	7260	48.63	8.16	9.39	1.85	14.77	17.20
Mid War	14377	2090	2544	612	4321	4618	50.34	7.32	8.91	2.14	15.13	16.17
Later War	5760	1238	972	194	1587	1668	50.44	10.84	8.51	1.70	13.90	14.61

	RetweetsCount			Proportion(%)		
sentiment	negative	neutral	positive	negative	neutral	positive
phase						
Early War	44816	15009	7556	66.51	22.27	11.21
Mid War	31994	10344	5877	66.36	21.45	12.19
Later War	243917	289714	121649	37.22	44.21	18.56

	RetweetsCount						Proportion(%)					
emotion	anger	fear	joy	love	sadness	surprise	anger	fear	joy	love	sadness	surprise
phase												
Early War	29254	7877	5797	801	9983	13669	43.42	11.69	8.60	1.19	14.82	20.29
Mid War	24003	4480	3912	581	6292	8947	49.78	9.29	8.11	1.21	13.05	18.56
Later War	168119	67017	116757	15878	67826	219683	25.66	10.23	17.82	2.42	10.35	33.53

	LikesCount			Proportion(%)		
sentiment	negative	neutral	positive	negative	neutral	positive
phase						
Early War	201248	82611	48518	60.55	24.85	14.60
Mid War	162543	54672	37919	63.71	21.43	14.86
Later War	24162	62382	4344	26.58	68.64	4.78

	LikesCount						Proportion(%)					
emotion	anger	fear	joy	love	sadness	surprise	anger	fear	joy	love	sadness	surprise
phase												
Early War	145919	25419	35247	4361	44522	76909	43.90	7.65	10.60	1.31	13.40	23.14
Mid War	124951	16805	26316	3653	36909	46500	48.97	6.59	10.31	1.43	14.47	18.23
Later War	15169	30702	5134	602	6971	32310	16.69	33.78	5.65	0.66	7.67	35.55

4.4 Quality Assurance

To ensure the quality and reliability of the project, several measures were adopted throughout development:

- Proper validation of datasets before processing
- Consistent preprocessing techniques across all phases
- Verification of model outputs using sample test cases
- Use of structured coding practices to reduce errors
- Cross-checking of graphical outputs for correctness

Additionally, the system was tested under different input conditions to ensure robustness and scalability. The final outputs were reviewed to confirm that they align with the project objectives and expected analytical outcomes.

Chapter 5

Standards Adopted

5.1 Design Standards

The design of the proposed system follows established software engineering principles to ensure clarity, scalability, and maintainability. Standard design practices were adopted to create a structured and modular architecture.

The system design is influenced by guidelines from internationally accepted standards such as **IEEE software design practices**. The architecture is divided into multiple functional modules including data collection, preprocessing, sentiment analysis, emotion detection, engagement analysis, and visualization.

To represent system structure and workflow, **Unified Modeling Language (UML)** concepts and block diagrams are used. These diagrams help in clearly illustrating the interaction between different components of the system.

Additionally, the design ensures:

- **Modularity**, allowing independent development and testing of components
- **Scalability**, enabling the system to handle larger datasets in future
- **Reusability**, so that modules can be extended or reused in similar applications

The overall design approach ensures that the system remains flexible and easy to maintain.

5.2 Coding Standards

Proper coding standards were followed to ensure readability, consistency, and efficiency of the implementation. The code was written in a structured manner using best programming practices.

Key coding standards followed include:

- Use of **meaningful and descriptive variable names** for better understanding
- Maintaining **consistent indentation and formatting** throughout the code
- Breaking the program into **small, modular functions**, each performing a specific task
- Avoiding redundant code by promoting **code reusability**
- Including **comments and documentation** to explain important sections
- Ensuring proper handling of **exceptions and errors**

- Following **logical flow and clean structure** for easy debugging and testing

These practices improve code maintainability and make it easier for future enhancements.

5.3 Testing Standards

Testing and validation of the system were conducted following standard quality assurance practices to ensure accuracy and reliability of results.

The following testing strategies were applied:

- **Unit Testing**: Individual components such as preprocessing, sentiment classification, and emotion detection were tested separately
- **Integration Testing**: Combined modules were tested to ensure smooth data flow between different stages
- **Functional Testing**: Verified that the system meets all specified functional requirements
- **Output Validation**: Results were cross-checked using sample inputs to ensure correctness

The system was also tested using multiple datasets from different phases to ensure consistency and robustness.

Additionally, test cases were designed to evaluate:

- Accuracy of sentiment classification
- Correctness of emotion detection
- Reliability of engagement metrics
- Validity of temporal analysis outputs

By following these standards, the system ensures high reliability, correctness, and performance.

Chapter 6

Conclusion and Future Scope

.6.1 Conclusion

This project presents a **multilingual emotion-aware sentiment intelligence approach** for analyzing public discourse during war situations. The system integrates sentiment analysis, emotion detection, and engagement metrics to examine how public reactions evolve across early, mid, and later phases of conflict.

The results demonstrate that public sentiment varies over time. The early phase shows higher uncertainty and emotional intensity, the mid phase reflects mixed and structured discussions, while the later phase indicates stabilization and adaptation. These findings highlight the importance of temporal analysis in understanding dynamic events.

Emotion detection provides deeper insights beyond basic sentiment classification, and engagement metrics such as likes and retweets help measure the impact and spread of opinions. Overall, the proposed system supports multilingual data, captures emotional variations, and produces interpretable visual outputs. The project successfully achieves its objective of analyzing temporal evolution of public discourse.

6.2 Future Scope

The system can be further improved by integrating real-time twitter data streaming to analyze live social media content during ongoing events. This would enhance the practical applicability of the framework.

Additional improvements may include sarcasm and irony detection to better interpret complex textual expressions, as well as improved noise handling for informal and highly unstructured social media text. Incorporating multimodal inputs can also provide richer contextual understanding.

The framework can also be extended to other domains such as disaster monitoring, market sentiment analysis, and public health campaigns, making it a versatile tool for large-scale social data analysis.

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INDIVIDUAL CONTRIBUTION REPORT:

MULTILINGUAL EMOTION-AWARE SENTIMENT INTELLIGENCE SYSTEM FOR TEMPORAL ANALYSIS OF PUBLIC DISCOURSE ACROSS WAR PHASES

RAJTANU DASGUPTA
2206200

Abstract: This project focuses on analyzing multilingual public discourse during war phases using sentiment and emotion-aware techniques. The objective is to understand how emotions and opinions evolve over time using computational methods.

Individual contribution and findings: I was primarily responsible for data collection and preprocessing. I worked with datasets obtained from social media sources and handled tasks such as cleaning noisy text, removing duplicates, and preparing structured datasets for analysis. I also contributed to handling multilingual text by applying normalization and tokenization techniques.

During implementation, I realized that preprocessing plays a crucial role in improving model performance. Even small improvements in cleaning significantly affected sentiment accuracy. I also explored challenges related to inconsistent language usage and informal expressions.

Individual contribution to project report preparation: I contributed mainly to Chapter 1 (Introduction) and Chapter 3 (Problem Statement and SRS).

Individual contribution for project presentation and demonstration: I explained data collection, preprocessing pipeline, and problem definition during the project presentation.

Full Signature of Supervisor: ……………………………….

Full signature of the student: ………………………………..

INDIVIDUAL CONTRIBUTION REPORT:

MULTILINGUAL EMOTION-AWARE SENTIMENT INTELLIGENCE SYSTEM FOR TEMPORAL ANALYSIS OF PUBLIC DISCOURSE ACROSS WAR PHASES

RICHIK DEY
2206203

Abstract: The project aims to provide a temporal analysis of sentiment and emotions across different war phases using machine learning and deep learning techniques.

Individual contribution and findings: I worked on model development and sentiment classification. I implemented machine learning and deep leaning models to classify text into positive, negative, and neutral categories. I also contributed to tuning parameters and evaluating model performance.

One key learning was understanding the trade-off between model complexity and performance. Simpler models performed well after proper preprocessing. I also observed how sentiment distribution changes across different time phases.

Individual contribution to project report preparation: I contributed to Chapter 4 (Methodology and Implementation).

Individual contribution for project presentation and demonstration: I demonstrated model working, classification logic, and evaluation results.

Full Signature of Supervisor: ……………………………….

Full signature of the student: ………………………………..

INDIVIDUAL CONTRIBUTION REPORT:

MULTILINGUAL EMOTION-AWARE SENTIMENT INTELLIGENCE SYSTEM FOR TEMPORAL ANALYSIS OF PUBLIC DISCOURSE ACROSS WAR PHASES

SOUMYA BHATTACHARJEE
2206219

Abstract: This project explores emotion-aware sentiment analysis using multilingual datasets to understand public reactions during different stages of war.

Individual contribution and findings: My primary responsibility was emotion detection and analysis. I implemented techniques to classify emotions such as anger, fear, sadness, and joy. I also worked on generating visualizations like graphs and charts to represent emotional trends.

I found that emotion analysis provides deeper insights compared to basic sentiment analysis. For example, two negative sentiments may represent completely different emotional states.

Individual contribution to project report preparation: I contributed to Chapter 2 (Literature Review) and Chapter 4 (Result Analysis).

Individual contribution for project presentation and demonstration: I explained emotion detection models and graphical outputs.

Full Signature of Supervisor: Full signature of the student:

……………………………. ……………………………..

INDIVIDUAL CONTRIBUTION REPORT:

MULTILINGUAL EMOTION-AWARE SENTIMENT INTELLIGENCE SYSTEM FOR TEMPORAL ANALYSIS OF PUBLIC DISCOURSE ACROSS WAR PHASES

SUBHADEEP MONDAL
2206221

Abstract: The project provides a complete analytical framework combining sentiment, emotion, and temporal insights for public discourse analysis.

Individual contribution and findings: I worked on temporal analysis and integration of early, mid, and later war phases. I analyzed how sentiment and emotions evolve across time and helped in combining results into a unified framework.

I also contributed to system integration and final testing. I observed that temporal segmentation significantly improves interpretability of results.

Individual contribution to project report preparation: I contributed to Chapter 5 (Standards) and Chapter 6 (Conclusion).

Individual contribution for project presentation and demonstration: I explained temporal analysis and overall system workflow.

Full Signature of Supervisor: ……………………………….

Full signature of the student: ………………………………..

TURNITIN PLAGIARISM REPORT

"MULTILINGUAL EMOTION-AWARE SENTIMENT INTELLIGENCE SYSTEM FOR TEMPORAL ANALYSIS OF PUBLIC DISCOURSE"

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